MONISHA GNANASEKAR

Design Engineer Aspirant | SolidWorks & AutoCAD | 3D CAD Modeling | Assembly & Drafting

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SUMMARY

Agricultural Engineering student with hands-on experience in CAD modeling, mechanical design, and prototyping. Skilled in SolidWorks and AutoCAD, with experience creating 3D models, assemblies, and engineering drawings, including CSWA-level design exercises. Worked on the WeedX precision farming robot project, gaining practical exposure to product development and engineering problem-solving. Interested in mechanical design, product development, and innovative engineering solutions.

EXPERIENCE

Tamirabot Advanced Engineering **APR 2025 - JAN 2026**

Project Trainee

- Contributed to early-stage prototyping of an agri-robotic weed detection system as part of a small engineering team
- Developed and tested embedded control logic on Arduino for sensor interfacing and basic actuation
- Assisted in mechanical assembly of the robot chassis, advancing the prototype to a working hardware stage with sensors operational for testing

TOSET Land Survey Institute **NOV 2024 - JAN 2025**

Survey Trainee Intern

- Completed a short-term hands-on training covering land surveying techniques and drafting workflows
- Operated Total Station and Auto Level instruments for field measurements, elevation readings, and basic land survey data collection
- Used AutoCAD to draft and plot survey data into site layout and contour drawings
- Gained practical exposure to translating field survey readings into accurate technical drawings, bridging fieldwork and CAD-based documentation

EDUCATION

2023 - 2027 **B.Tech Agricultural Engineering**

SSCET, SALEM CGPA: 8.32

SKILLS

CAD & Design Skills

1. SolidWorks
2. AutoCAD
3. 3D Part Modeling
4. Assembly Modeling
5. Engineering Drawing & Drafting
6. Sheet Metal Design

Technical Skills

1. Design Fundamentals
2. 2D to 3D CAD Conversion
3. Product Prototyping
5. Basic C Programming
6. Basic PythonSensor Integration

PROJECTS

- **Nalam Soil –** Developed an IoT-based smart fertilizer advisory system using an NPK sensor and Arduino Uno R4 WiFi for real-time soil nutrient monitoring. Delivers site-specific fertilizer recommendations using TNAU/ICAR guidelines to reduce overuse and improve crop nutrition precision.
- **FarmSync –** Designed a concept for a digital agriculture platform connecting farmers with surveyors and agricultural planning services to support land management and climate-smart farming practices.
- **WeedX (Next-Gen Precision Farming Robot) -** Developed a prototype precision farming robot for weed detection and removal using embedded systems and sensor integration. Participated in hardware assembly, testing, and field validation to reduce manual weeding requirements.

ACHIEVEMENTS

- StartupTN Smart Card Recipient for FarmSync AgriTech Platform.
- Presented research and project work at National and International Symposiums on Sustainable Agriculture.
- Completed a 12-Month Mentorship Program under V. Irai Anbu.

CERTIFICATIONS

NPTEL – Irrigation and Drainage Engineering (72%)

NPTEL – Modern Food Packaging Technologies (Elite, 80%)

DECLARATION

I hereby declare that the above-mentioned information is true to the best of my knowledge and belief.